Xiangling Xu 许湘灵

PhD student, PhIQuS, Inria Saclay

My Website
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Research Interests

I am interested in the capabilities and limits of quantum physics. To explore these questions, I study correlations in quantum networks, certifiable properties of quantum many-body systems, and cryptographic approaches to testing quantum devices. My main mathematical tools are the theory of operator algebras and noncommutative polynomial optimization.

Education

Ph.D. in Computer Science, Institut Polytechnique de Paris — 2023 – Present
Expected completion: 2027-05; supervisor: Marc-Olivier Renou.

M.Sc. Mathematics, ETH Zürich — 2020 – 2023
Thesis: *Quantum Nonlocality in Bilocal Networks: An Operator Algebraic Perspective*.
GPA: 5.69/6.00.

B.Sc. Mathematics and Physics, University of Toronto — 2016 – 2020
Mathematics and Physics Specialist Program; GPA: 3.95/4.00.

Publications and Preprints

1. **An operational characterization of finite-dimensional quantum theory**.
L. J. Dmello, X. Xu, M.-O. Renou, and D. Gross.
2026-10. arXiv: 2610.03694 [quant-ph].

2. **Quantum inflation and source-transfer hierarchies are complete for tree networks**.
X. Xu, I. Klep, and M.-O. Renou.
2026-09. arXiv: 2609.19310 [quant-ph].

3. **Fermions are fundamentally more nonlocal than Bosons**.
F. M. Kalarde, S. Boreiri, X. Xu, L. Tendick, S. Beigi, P. Perinotti, T. Guaita, and M.-O. Renou.
2026-06. arXiv: 2606.12363 [quant-ph].

4. **The bulk spectral gap is certifiable from above but uncomputable from below**.
X. Xu, M. Schötz, J. Wang, V. Magron, I. Klep, O. Fawzi, and M.-O. Renou.
2026-06. arXiv: 2606.03836 [quant-ph].

5. **Comment on "Quantum theory based on real numbers cannot be experimentally falsified": on the compatibility of physical principles with information theory for fermions**.
F. M. Kalarde, X. Xu, and M.-O. Renou.
2026-04. arXiv: 2604.07425 [quant-ph].

6 **Two convergent NPA-like hierarchies for the quantum bilocal scenario**.
M.-O. Renou, X. Xu, and L. T. Ligthart.
Journal of Mathematical Physics, vol. 67, no. 1, Art. no. 012203, 2026-01. DOI: 10.1063/5.0211008

7 **Quantitative quantum soundness for all multipartite compiled nonlocal games**.
M. Baroni, I. Klep, D. Leichtle, M.-O. Renou, I. Šupić, L. Tendick, and X. Xu.
2025-09. arXiv: 2509.25145 [quant-ph]. QIP 2026 (contributed talk); alphabetical author order.

8 **Quantitative quantum soundness for bipartite compiled Bell games via the sequential NPA hierarchy**.
I. Klep, C. Paddock, M.-O. Renou, S. Schmidt, L. Tendick, X. Xu, and Y. Zhao.
2025-07. arXiv: 2507.17006 [quant-ph]. QIP 2026 (contributed talk); alphabetical author order.

9 **Quantitative Tsirelson's theorems via approximate Schur's lemma and probabilistic Stampfli's theorems**.
X. Xu, M.-O. Renou, and I. Klep.
2025-05. arXiv: 2505.22309 [quant-ph].

10 **On the kinematic morphology around haloes**.
X. Luo, X. Xu, and X. Wang.
Monthly Notices of the Royal Astronomical Society, vol. 518, no. 4, pp. 6059–6064, 2023. DOI: 10.1093/mnras/stac3500

11 **Quantum nonlocality in bilocal networks: an operator algebraic perspective**.
X. Xu.
M.S. thesis, ETH Zurich, 2023. DOI: 10.3929/ethz-b-000613670

Selected Talks

The bulk spectral gap is RE-complete: certifiable from above, uncomputable from below

2026-11 Journées Informatique Quantique, Toulouse, France (contributed).
4th Colloquium GDR TeQ, Gif-sur-Yvette, France (contributed).

Quantitative quantum soundness for compiled nonlocal games and the sequential NPA hierarchy

2026-08 QCrypt, Ottawa, Canada (contributed).
2026-06 SIAM Conference on Optimization (OP26), Edinburgh, United Kingdom (invited).
2026-01 QIP 2026, Riga, Latvia (two contributed talks: bipartite and multipartite games).
2025-12 Quantum Meets, online (invited).
2025-11 3rd Colloquium GDR TeQ, Grenoble, France (contributed).
2025-10 YQIS25, Barcelona, Spain (contributed).
2025-07 IWOTA, Enschede, Netherlands (invited).
2025-06 IQC-PCQT-Quantum Saclay Workshop, Paris, France (invited).

Characterizing quantum bilocal networks with generalized NPA hierarchies

2024-09 CausalWorlds, Waterloo, Canada (contributed).
2023-11 Defi EQIP Workshop, Lyon, France (contributed).

Selected Talks (continued)

2023-10 Saclay Quantum Seminar, Palaiseau, France (invited).

Other research seminars

2026-09 Quriosity, Inria / Télécom Paris / IP Paris.
2026-08 Quantum Science Center of Guangdong-Hong Kong-Macao Greater Bay Area (Ranyiliu Chen).
2026-07 Nanjing University (Penghui Yao).
2026-02 Institut de physique théorique, CEA (Jean-Daniel Bancal).
2025-12 AMSS, Chinese Academy of Sciences (Jie Wang).
2025-11 EPFL (Thomas Vidick).
2025-10 Freie Universität Berlin (Jens Eisert).
2025-02 QAT team, ENS Paris.
2024-07 University of Waterloo (William Slofstra).
2023-12 University of Cologne (David Gross).

Posters

2026-08 CQCC & GQSF, Shenzhen, China.
2024-11 2nd Colloquium GDR TeQ, Paris, France.
YQIS24, Paris, France.

Research Visits

2026-10 – 2026-11 David Gross — University of Cologne.
2026-07 Penghui Yao — Nanjing University.
2026-05, 2025-12 Jie Wang — AMSS, Chinese Academy of Sciences.
2025-10 – 2025-11 Thomas Vidick — EPFL.
2024-09 Igor Klep — University of Ljubljana.
2024-07 William Slofstra — Institute for Quantum Computing, University of Waterloo.
2023-12 David Gross — University of Cologne.

Academic Service

Organization

2024-02 – Present Co-organizer, COMPUTE (QuantERA) project activities.
2025-03 – 2026-03 Co-organizer, Seminars in Commutative and Noncommutative Polynomial Optimization for Quantum Theory and Gaspard Monge Visiting professor program for Igor Klep.
2025-03 Co-organizer, 828th WE-Heraeus Seminar on Operator Theory and Polynomial Optimization in Quantum Information Theory, Bad Honnef.
2023-09 Co-organizer, Saclay Quantum Seminar on the NPA Hierarchy.

Academic Service (continued)

Reviewing

Quantum; *Operator Theory* (Springer, chapter); QIP and CausalWorlds.

Selected Awards and Funding

2026 Erasmus+ Mobility Funding, research visit to the University of Cologne.

2024 Erasmus+ Mobility Funding, research visit to the University of Ljubljana.

2020 McNab Undergraduate Scholarship, University of Toronto.

2018 Class of 3To and Associates Scholarship in Mathematics and Physics, University of Toronto.

2017 Summer Undergraduate Research Program, Canadian Institute for Theoretical Astrophysics.

2016 President's Scholar of Excellence Program, University of Toronto.

Teaching Experience

2024, 2025 Lecturer, *Recent Trends in Parallel, Distributed, and Quantum Computing*, QDCS Master's program, Université Paris-Saclay.

2022 – 2023 Teaching assistant, *Linear Algebra I & II*, ETH Zürich.

2021 – 2022 Teaching assistant, *Analysis I & II*, ETH Zürich.

Skills

Computing Python, MATLAB, C, Mathematica; ncpol2sdpa, YALMIP.

Languages Chinese (native), English (fluent), German (beginner).